

"The trend is moving towards solid-state illumination"

With LEDs and lasers becoming more efficient and environment-friendly, light sources are changing in digital light processing (DLP)-based products. Kent Novak, senior vice president & general manager, DLP Products, Texas Instruments spoke to Pankaj Vashisht of EFY about the advancement of DLP in non-display applications and new trends in DLP light sources



KENT NOVAK
SENIOR VICE PRESIDENT &
GENERAL MANAGER, DLP
PRODUCTS, TEXAS INSTRUMENTS

Q. What enhancements would a product inherit, if it is designed to run with a modern DLP chip?

General trends show that over time, newer chips are developed with higher resolution, better brightness and light efficiency. Specifically, use of microscopically small mirrors as reflectors inside the chips and changing tilt angles of the mirrors have achieved more brightness. As compared to previous generation, systems designed using these new chips are 50% brighter using the same power on the LEDs, increasing their light efficiency. Engineers who are designing consumer applications where battery life is more important, get more brightness with less power consumed by using a modern DLP chip.

Q. Is there any limitation on the size or resolution of the image while using DLP technology?

The size of the image mainly depends on the projector brightness, optics configuration and the distance between the projector and display surface or projector screen. DLP projectors can illuminate screen sizes of 25.4cm (10-inch) width to 30.5m (100-feet) width. The former is used in pico phones and the latter in cinema halls.

Q. What benefits can an engineer get by switching his current design to the newer DLP technology?

Faster switching speeds with DLP gives it an edge over other existing technologies. Whether it is cinema projection, 3D projection or 3D inspection, or spectroscopy, faster moving mirrors give the advantage of faster switching speed which, in turn, allows those applications to go faster. This imaging speed of DLP chipsets, when combined with consumer applications like digital cameras and projectors, results in highly reliable and accurate systems.

Q. How should a designer go about DLP-based product designing?

The DLP chip, or digital micro-mirror device (DMD), is an array of micro-mirrors that can be used for high-speed, efficient and reliable light steering. DLP technology can be considered in any application where light is steered or manipulated. Development tools and reference designs are available for engineers for ensuring a faster time to market.

Q. What advantages can DLP present for industrial, medical or automotive sectors?

In the industrial or automotive space, the traditional expectations are excellent colours, contrast and brightness. For example, in head-up displays for automobiles, the common expectation is to have a wider and wider field of view with good brightness. You can have that with DLP technology while existing technologies can not do that. Medical applications like DLP-enabled hyperspectral imaging method opens up a wide range of possibilities in spectroscopy. DLP provides imaging technology with real time mapping and visualisation data to assist surgeons with performing difficult procedures.

Q. What are the trends in light sources for DLP-based products?

With the emergence of solid-state lighting, light sources have gained importance in DLP-based products. Many of the projectors today, whether cinema projectors or class-room projectors, still use bulb as a light source. But all pico projectors use LEDs because the most cost-effective solution to get more lumens is LED. All automotive solutions use LEDs as they fit best for the brightness that we need. Then there are laser-based projectors. The trend is moving towards solid-state illumination because more brightness per watt, or you can say brightness per rupee or dollar, is the main concern, and at the same time it is the right solution for the environment as well.

Q. How does DLP 3D projection really work?

DLP technology uses millions of microscopic digital mirrors that reflect light to create a stunning picture for the best projectors in the market. This imaging technology is so fast, it can actually produce two images on the screen at the same time—one for the left eye and one for the right eye. Then 3D glasses combine the two images to create an amazing 3D effect. DLP has enabled a single projector box to be able to display either 2D or 3D content simply depending on the content being run through it. This can greatly benefit classrooms, as they can future-proof projector purchase. They can purchase a 3D projector today, for no extra cost, and purchase 3D glasses when they are ready to beam 3D content. ●